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LABELS AND ANNOTATIONS

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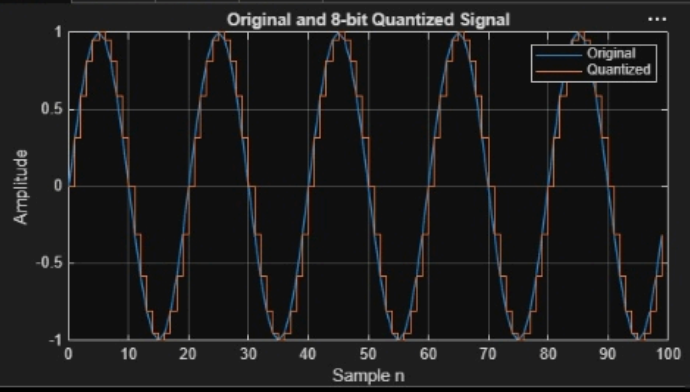
ip.m | exp4dsp.m | exp5dsp.m | exp6dsp.m | exp7dsp.m | exp8dsp.m | exp9dsp.m

/MATLAB Drive/exp9dsp.m

```
1   clc; clear; close all;
2
3   %% PARAMETERS
4   A=1; f=50; Fs=1000; N=1000;
5   n=0:N-1; x=A*sin(2*pi*f*n/Fs);
6   B=8; bits=[4 8 12 16];
7
8   %% 8-BIT QUANTIZATION
9   D=2*A/2^B;
10
11  xr=D*round(x/D); % Rounding
12  xt=D*fix(x/D); % Truncation
13
14  er=x-xr; % Rounding error
15  et=x-xt; % Truncation error
16
17  SQr=10*log10(mean(x.^2)/mean(er.^2));
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Figures

Figure 1 | Figure 2 | Figure 3 | Figure 4



Command Window

```
========== FINAL RESULT ==========
Uniform quantization using rounding and truncation was performed.
Quantization error and SQNR were calculated.
Increasing word length reduces quantization error and improves SQNR.
Finite word-length effects are therefore verified.
>>
```

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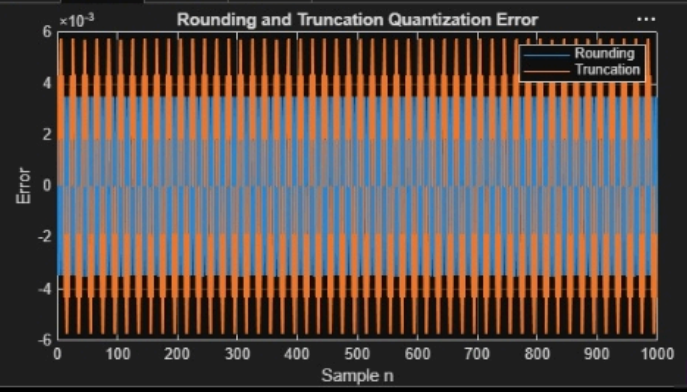
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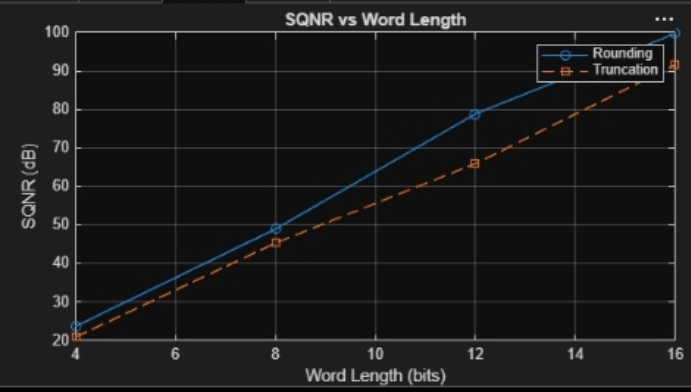
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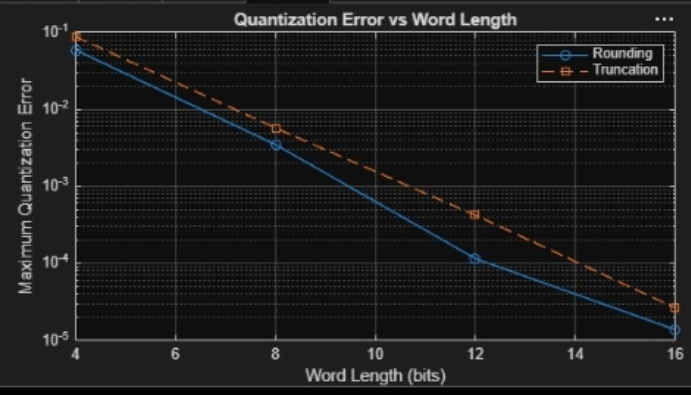
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